

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress
(256-bit S, K, T, r, C, TID)

Stage 1: Ingress Registration & TID Queuing (1 Cycle)

AXI4-Stream 256-bit Ingress
{S, K, T, r, C_mkt, TID}
Direct Single-Cycle Ingress
Zero Serial Divider at Ingress

Zero-BRAM Context Store

64 x 32-bit Distributed LUTRAM
SRL32 Shift Register Delay Lines
Zero RAMB36/18 Consumed
Preserves BRAM for 10GbE / PCIe

Stage 2: Analytical Seeder (Brenner-Subrahmanyam, 64 Cyc)

$\sigma_0 = (2.5066/\sqrt{T}) * C / ((S+K)/2)$
Shared Digit-Recurrence $\sqrt{T}$ (29 Cyc)
Internal Non-Restoring Divider (33 Cyc)
Forwards $\sqrt{T}$ to Main Datapath

Stage 3: Core BS Datapath (126 Cycles, II=1)

Scale-Invariant Padé $\ln(S/K)$ (33 Cyc)
Matched 33-Cyc CORDIC Log Fallback
Horner 5th-Order Poly CDF $N(d)$
Single-Slot II=1 Loopback Ingestion

Stages 4 & 5: NR Update & 128-bit Egress (35 Cyc)

Stage 4: NR Divider (33 Cyc)
 $\Delta\sigma = \Delta C / \text{Vega}$
Parallel Gamma Divider
Stage 5: Egress Arbiter (2 Cyc)
Round-Robin 4-Core Complete
[127:0] = {TID, Gamma, Vega, Delta, sigma}

AXI4-Stream Egress
(128-bit sigma + Greeks)